

ASTRO BOT

An Intelligent Conversational AI Assistant

Multimodal · Retrieval-Augmented Generation · Workflow Integration

01 PROBLEM DEFINITION

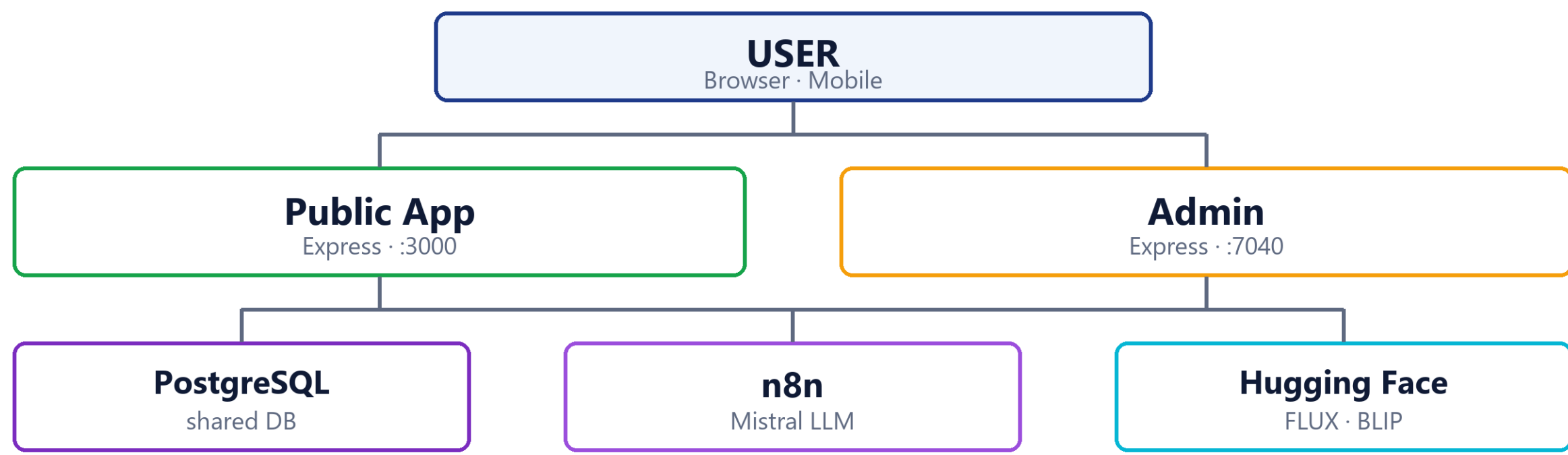
Modern AI assistants are scattered across separate tools — one product for chat, another for image generation, another for document search, another for OCR and audio. Users juggle several paid subscriptions to reach state-of-the-art models, while self-hosting stays complex. AstroBot fills that gap: one free-to-use web application that unifies conversational AI, multimodal input, document Q&A and reusable workflows behind a clean chat interface — plus an admin dashboard for operations.

02 PROJECT OBJECTIVES

- ✓ A unified, containerised conversational AI assistant
- ✓ Mistral LLM routed through an n8n visual workflow
- ✓ Image generation (FLUX.1) + image-to-image editing
- ✓ Document Q&A via RAG with MiniLM embeddings
- ✓ OCR (Tesseract) + audio transcription (Whisper)
- ✓ On-demand PDF report generation from chat
- ✓ Admin dashboard with live stats & user management

03 ARCHITECTURE

Three Docker services on a private network, sharing one PostgreSQL

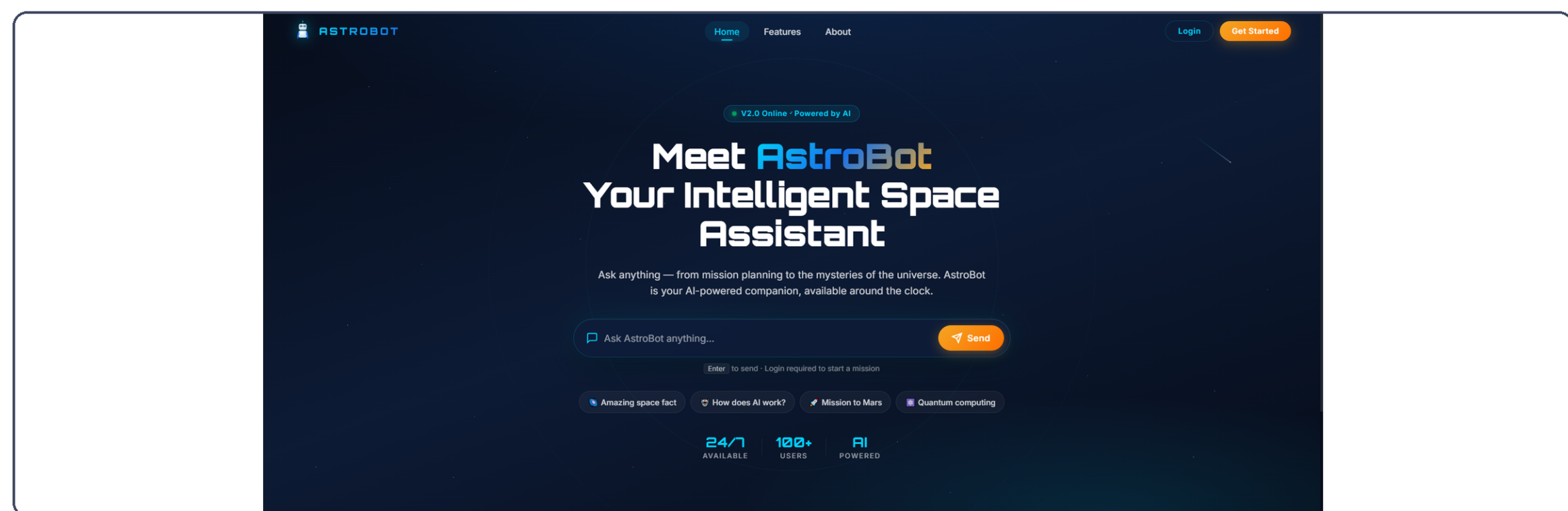


04 KEY FEATURES

- **Conversational chat**
Mistral medium via n8n
- **Image generation**
FLUX.1-schnell + fal-ai
- **PDF reports**
PDFKit, styled output
- **Document Q&A — RAG**
MiniLM-L6-v2 embeddings
- **OCR + captioning**
Tesseract.js + BLIP
- **Audio transcription**
Groq Whisper-large-v3
- **Admin dashboard**
Stats · users · audit
- **Notebook mode**
Pin a PDF as context

05 TRY ASTROBOT IN ACTION

The public landing page — ask the bot anything from the central chat bar



Central chat bar · Animated mascot · 10 languages · Dark / light

08 TRY IT LIVE

Scan the QR code to launch AstroBot in your browser



Scan to launch AstroBot
<http://76.13.62.195:3000>

06 PROJECT TEAM

Four developers — Computer Engineering, Faculty of Engineering



Tidiane Konaté

Architecture · Backend · Deployment



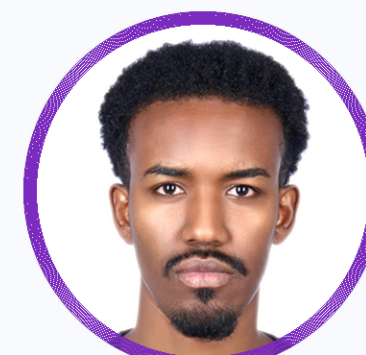
Sidi Mohamed Sall

Vision · Product · UX



Ahmed Essalem

AI · LLM · Multimodal



Saad Ibrahim Houssein

QA · Frontend · Testing

07 ADVISOR



Assist. Prof. Dr. Yücel TEKİN

Project Supervisor

Source code
github.com/tkonate788/astrobot

Built with
Node.js · Express · PostgreSQL · n8n · Mistral · Hugging Face · Docker